

Problem Set 7

1. FIR filter classification & frequency response derivation

Given four causal discrete-time FIR systems with impulse responses defined for $0 \leq n \leq M$:

1. $h_1[n] = [1, 3, 5, 3, 1]$

2. $h_2[n] = [2, -1, 4, 4, -1, 2]$

3. $h_3[n] = [1, 2, 0, -2, -1]$

4. $h_4[n] = [-1, 3, -3, 1]$

(a) Classify each system into its corresponding FIR Type (I, II, III, or IV). Specify the filter order M and sequence length N for each.

(b) For $h_2[n]$, derive the compact closed-form expression for its frequency response $H_2(e^{j\omega})$. Express $H_2(e^{j\omega})$ in the form $A(\omega)e^{-jM/2}$ where $A(\omega)$ is a real-valued trigonometric sum.

(c) For $h_3[n]$, derive $H_3(e^{j\omega})$ and show that it exhibits a constant 90° ($\pi/2$ rad) phase shift relative to the linear phase delay term $e^{-jM/2}$.

2. Zero Constellation Symmetry & System Synthesis

(a) Prove that if $z_0 = re^{j\theta}$ (with $r \neq 1$ and $\theta \neq 0, \pi$) is a zero of a causal, real-coefficient linear-phase FIR filter $H(z)$, then $z_0, z_0^*, 1/z_0$, and $1/z_0^*$ must also be zeros of $H(z)$ with equal multiplicity.

(b) A Type I FIR filter of order $M = 4$ ($N = 5$) with real impulse response $h[n]$ has a complex zero at:

$$z_1 = \frac{1}{2}e^{j\pi/3}$$

- List all four zeros of $H(z)$ in both polar and Cartesian forms.
- Derive the normalized system transfer function $H(z)$ given that $H(e^{j0}) = 1$.
- Determine the explicit impulse response sequence $h[n]$ for $n = 0, 1, 2, 3, 4$.

3. System interconnections and composite types

Consider two causal linear-phase FIR filters: $H_A(z)$ of Type II with order M_A , and $H_B(z)$ of Type IV with order M_B .

(a) Determine the FIR Type (I, II, III, or IV) of the cascaded system $H_C(z) = H_A(z) \cdot H_B(z)$.

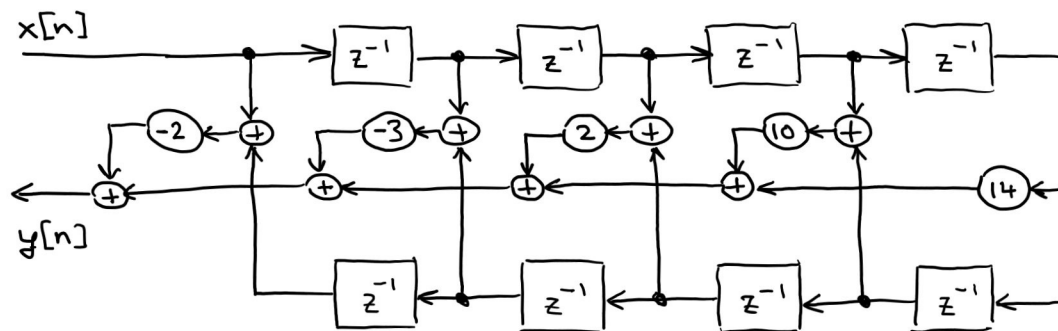
Express the overall order M_C in terms of M_A and M_B , and justify the symmetry of $h_C[n]$.

(b) Assume $M_A = M_B = M$. Determine whether the parallel interconnection $H_D(z) = H_A(z) + H_B(z)$ maintains strict generalized linear phase. Support your answer mathematically by evaluating the phase of $H_D(e^{j\omega})$.

(c) Can a parallel combination of a Type I filter $H_1(z)$ and a Type III filter $H_3(z)$ with identical delay $\tau = M/2$ yield a linear phase system? Explain.

4. Filter flowgraph (ECEN 5632 Problem)

Consider the filter flowgraph shown in the figure below:



The z^{-1} cells are delay cells and the bubbles with numbers in them are multipliers with the numbers giving the multiplication factors. The bubbles with + signs are adders, the branch labeled $x[n]$ is the input of the filter and the branch labeled $y[n]$ is the output of the filter.

(a) Determine the unit impulse response $h[n]$ of the filter shown above and display the result in the form of a labeled stem plot. What type of a filter is it?

(b) What would be the main advantage of implementing a filter in this way? Are there any disadvantages?

(c) Could you implement a Type III and/or Type IV FIR filter in this way? If yes, what would you have to change, if no, why not?